Reinforcement Learning Summative Assignment Report

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Video Recording: [Link to Video - Camera On, Full Screen]

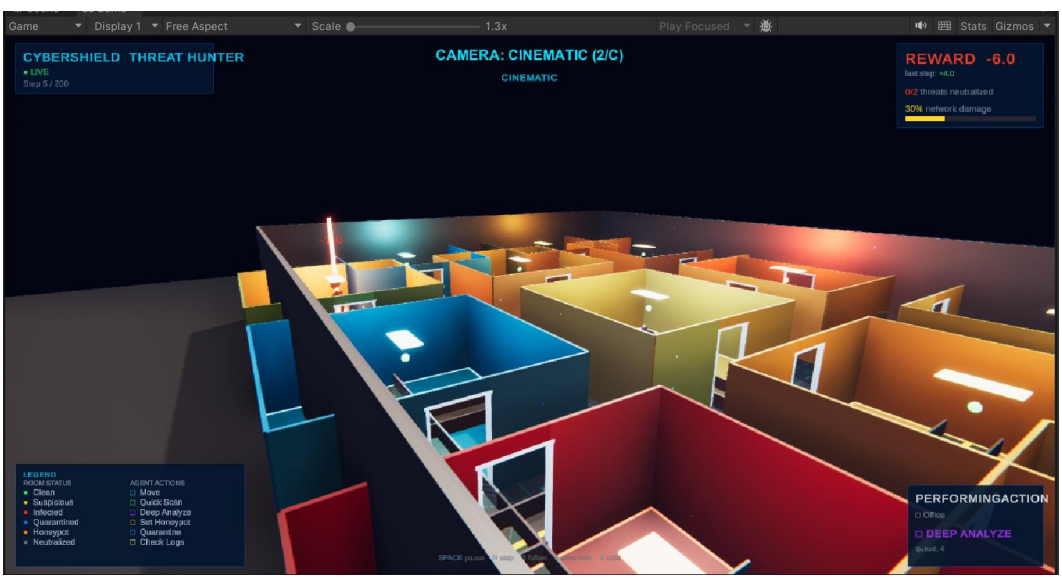
GitHub Repository: [Link to Repository]

1. Project Overview

This project implements a reinforcement learning agent for autonomous cyber threat hunting across a simulated enterprise network. The CyberShield Threat Hunter agent navigates a 14-node network topology containing hidden threats (malware, backdoors, cryptominers, and data exfiltration) and must discover and neutralize them before cumulative damage exceeds a critical threshold. Four RL algorithms — DQN, REINFORCE, PPO, and A2C — were trained and compared across 48 hyperparameter configurations (12 per algorithm). A Flask REST API demonstrates production integration, and the environment is visualized in Unity 3D via a TCP socket bridge.

2. Environment Description

Unity 3D Visualization of the CyberShield Threat Hunter Environment



The simulated enterprise network rendered in Unity 3D (Cinematic camera view). Each room represents a network node with color-coded threat states: green (clean), yellow (suspicious), red (infected), blue (quarantined), orange (honeypot), grey (neutralized). The agent (green figure) navigates through corridors between rooms. The CyberHUD overlay displays real-time metrics including camera mode, current action, reward, and agent status.

Agent(s)

The agent represents an automated cybersecurity analyst ("Threat Hunter Drone") that moves between network nodes to investigate, analyze, and neutralize cyber threats. It must balance thoroughness with urgency (threats deal ongoing damage) and accuracy (false positives incur penalties).

Action Space

The action space is Discrete(6) with the following actions:

Action ID	Name	Description
0	MOVE	Move to the next adjacent node in the network graph
1	QUICK_SCAN	Fast shallow scan — reveals obvious threats (anomaly > 0.5)
2	DEEP_ANALYZE	Slow thorough analysis — reveals hidden threats at any level
3	SET_HONEYPOT	Deploys honeypot to lure and reveal threats over time
4	QUARANTINE	Isolates node, stops threat spread but causes downtime
5	CHECK_LOGS	Examines historical activity, boosts threat indicator on suspicious nodes

Observation Space

The observation space is a continuous Box(89,) vector: per-node features (14 nodes × 6 features = 84 dims) including CPU usage, memory usage, network anomaly score, time since last scan, threat indicator level, and quarantine status; plus 5 global features (agent position, time remaining, threats neutralized, total damage, active honeypots). All values normalized to [0, 1].

Reward Structure

Event	Reward	Rationale
Correctly quarantining a threat	+10	Primary objective completion
Discovering hidden threat (deep analyze)	+5	Information gathering
Detecting anomaly (quick scan)	+2	Efficient reconnaissance
False positive (quarantine clean node)	-5	Penalizes reckless behavior
Each timestep	-1	Creates urgency pressure
Undetected threat causes breach	-15	Failure penalty
All threats neutralized (bonus)	+25	Episode completion reward

Terminal conditions: (1) All threats neutralized (success), (2) Damage exceeds threshold of 100 (failure), (3) 100 timesteps reached (timeout).

3. System Analysis and Design

Deep Q-Network (DQN)

The DQN implementation uses Stable Baselines 3 with a target network for stable Q-value estimation and an experience replay buffer for breaking temporal correlations. The policy network is an MLP mapping the 89-dim observation to Q-values for 6 actions. Epsilon-greedy exploration decays from 1.0 to 0.05. The best configuration used a smaller replay buffer (10,000), forcing the agent to learn from recent relevant experiences.

REINFORCE (Vanilla Policy Gradient)

A custom PyTorch implementation since SB3 does not include vanilla REINFORCE. The policy network outputs a categorical distribution over actions via softmax. After each episode, discounted returns are computed with optional baseline subtraction (mean return normalization) to reduce variance. The policy is updated using the REINFORCE gradient estimator.

Proximal Policy Optimization (PPO)

PPO uses a clipped surrogate objective to prevent destructively large policy updates. The SB3 implementation uses separate value and policy heads sharing an MLP backbone. The clipping parameter (0.2) limits the ratio of new to old action probabilities. An entropy bonus (0.01) encourages exploration.

Advantage Actor-Critic (A2C)

A2C is a synchronous actor-critic method using the advantage function to reduce gradient variance compared to REINFORCE. The best configuration used lower gamma (0.9), suggesting the agent benefits from focusing on immediate rewards in this environment.

4. Hyperparameter Experiments

DQN Experiments

Learning Rate	Gamma	Buffer Size	Batch Size	Exploration	Net Arch	Mean Reward
1e-4	0.99	50,000	64	0.05/0.3	[128,128]	7.43
5e-4	0.99	50,000	64	0.05/0.3	[128,128]	-4.00
1e-5	0.99	50,000	64	0.05/0.3	[128,128]	-13.50
1e-4	0.90	50,000	64	0.05/0.3	[128,128]	-3.50
1e-4	0.999	50,000	64	0.05/0.3	[128,128]	9.30
1e-4	0.99	50,000	32	0.05/0.3	[128,128]	1.90
1e-4	0.99	50,000	128	0.05/0.3	[128,128]	4.60
1e-4	0.99	10,000	64	0.05/0.3	[128,128]	11.98*
1e-4	0.99	50,000	64	0.01/0.5	[128,128]	2.38
1e-4	0.99	50,000	64	0.10/0.1	[128,128]	8.20
1e-4	0.99	50,000	64	0.05/0.3	[256,256,128]	7.28
1e-4	0.99	50,000	64	0.05/0.3	[512,256]	6.20

Best config: dqn_small_buffer (marked *). A smaller replay buffer forces learning from recent, relevant transitions. Very low learning rate (1e-5) prevented learning within 50K timesteps. Higher gamma (0.999) outperformed lower (0.9), indicating long-horizon planning matters for the +25 completion bonus.

REINFORCE Experiments

Learning Rate	Gamma	Hidden Sizes	Baseline	Mean Reward	Best Eval
1e-3	0.99	(128,128)	Yes	-11.45	-10.50
1e-3	0.99	(128,128)	No	-12.20	-8.30
5e-3	0.99	(128,128)	Yes	-14.00	-12.20
1e-4	0.99	(128,128)	Yes	-11.95	-3.10
1e-3	0.90	(128,128)	Yes	-7.85*	-4.80
1e-3	0.999	(128,128)	Yes	-11.20	-6.70
1e-3	0.99	(64,64)	Yes	-9.80	-4.90
1e-3	0.99	(256,256,128)	Yes	-11.10	-12.80
1e-3	0.99	(512,256)	Yes	-12.00	-10.90

1e-3	0.99	(32,32)	Yes	-14.08	-10.90
5e-4	0.99	(128,128)	No	-13.85	-12.30
3e-4	0.995	(256,128)	Yes	-10.80	-9.50

Best config: reinforce_low_gamma (marked *). REINFORCE struggled across all configurations due to high gradient variance. Low gamma (0.9) reduces variance by discounting distant noisy returns. Only 500 episodes of training is insufficient for this complex environment.

PPO Experiments

LR	Gamma	N Steps	Batch	Epochs	Clip	Ent Coef	Arch	Mean Reward
3e-4	0.99	256	64	10	0.2	0.01	[128,128]	11.35
1e-3	0.99	256	64	10	0.2	0.01	[128,128]	13.90
1e-5	0.99	256	64	10	0.2	0.01	[128,128]	-14.05
3e-4	0.90	256	64	10	0.2	0.01	[128,128]	5.25
3e-4	0.99	256	64	10	0.3	0.01	[128,128]	1.55
3e-4	0.99	256	64	10	0.1	0.01	[128,128]	-1.65
3e-4	0.99	256	64	10	0.2	0.05	[128,128]	12.90
3e-4	0.99	256	64	10	0.2	0.00	[128,128]	8.25
3e-4	0.99	512	64	10	0.2	0.01	[128,128]	13.80
3e-4	0.99	256	64	10	0.2	0.01	[256,256,128]	14.30*
3e-4	0.99	256	64	3	0.2	0.01	[128,128]	-12.25
3e-4	0.99	256	128	10	0.2	0.01	[128,128]	4.65

Best config: ppo_deep_net (marked *). The deeper architecture (256-256-128) captures complex state-action relationships. PPO's clipping mechanism provides robustness to learning rate changes. Entropy regularization is critical: high entropy (+12.90) outperformed no entropy (+8.25). Too few epochs (3) caused severe underfitting (-12.25).

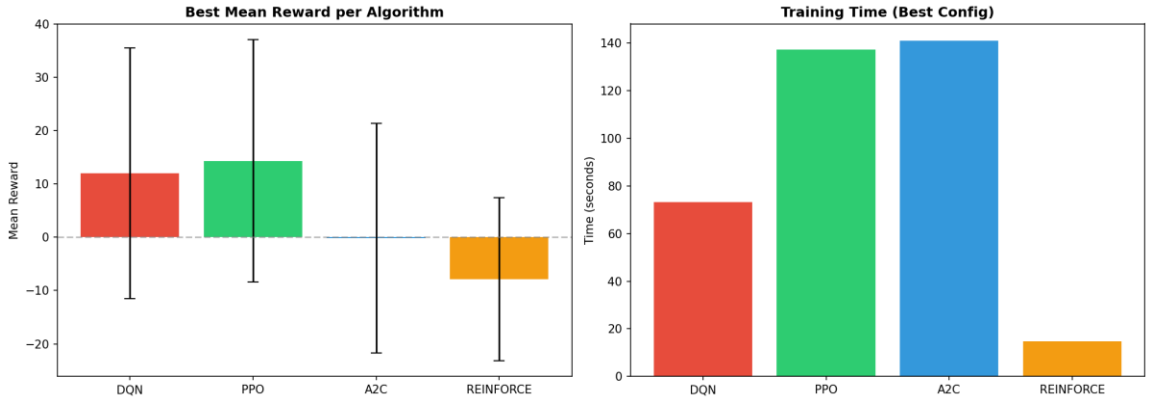
A2C Experiments

Learning Rate	Gamma	N Steps	Ent Coef	VF Coef	Net Arch	Mean Reward
7e-4	0.99	5	0.01	0.5	[128,128]	-2.10
1e-3	0.99	5	0.01	0.5	[128,128]	-0.75
1e-4	0.99	5	0.01	0.5	[128,128]	-12.20
7e-4	0.90	5	0.01	0.5	[128,128]	-0.18*
7e-4	0.999	5	0.01	0.5	[128,128]	-0.35
7e-4	0.99	20	0.01	0.5	[128,128]	-12.55
7e-4	0.99	5	0.05	0.5	[128,128]	-13.00
7e-4	0.99	5	0.00	0.5	[128,128]	-13.50
7e-4	0.99	5	0.01	1.0	[128,128]	-14.25
7e-4	0.99	5	0.01	0.5	[256,256,128]	-12.15
7e-4	0.99	5	0.01	0.5	[512,256]	-12.25
7e-4	0.99	10	0.01	0.5	[128,128]	-8.45

Best config: a2c_low_gamma (marked *). A2C was the weakest SB3 algorithm. Short rollouts (n_steps=5) work best; longer rollouts accumulate stale on-policy data. A2C is highly sensitive to entropy tuning and would benefit from significantly more training time.

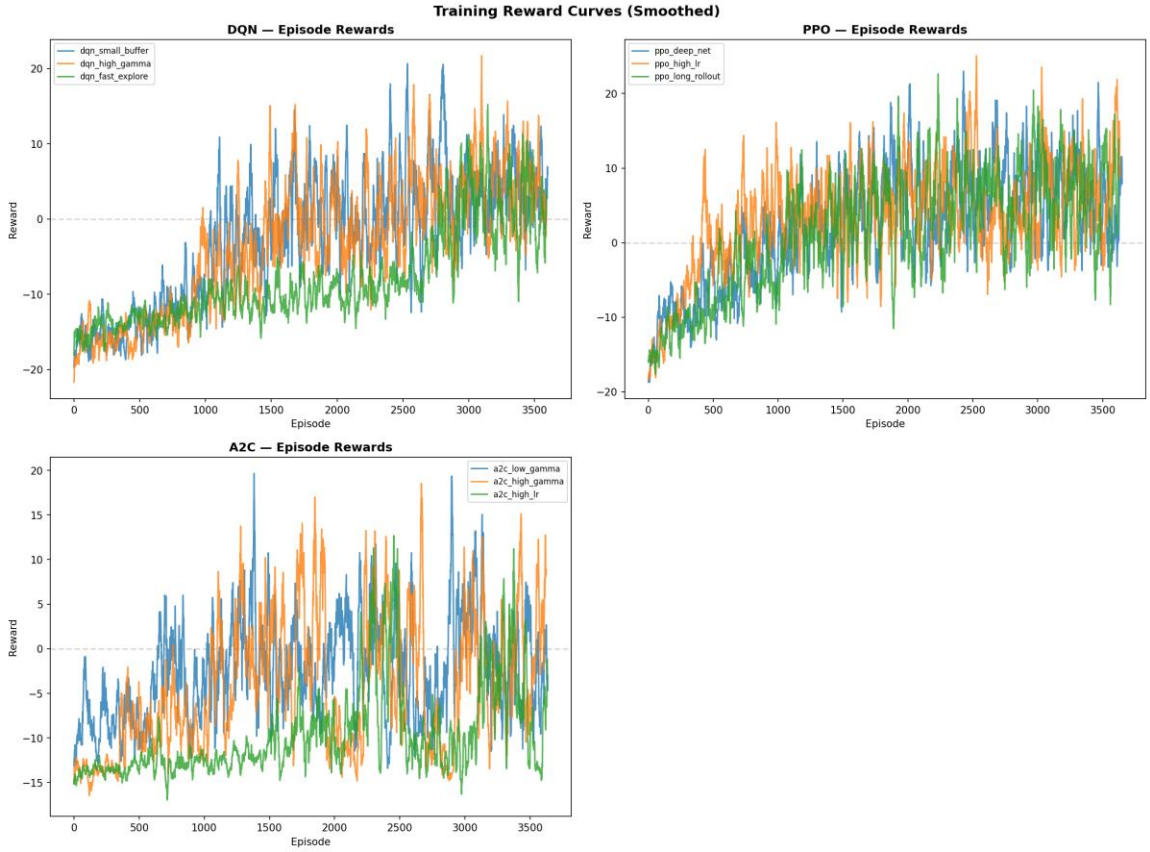
5. Results Discussion

Algorithm Comparison



PPO achieved the highest mean reward (+14.30), followed by DQN (+11.98), A2C (-0.18), and REINFORCE (-7.85). PPO and DQN both learned effective threat hunting policies, while A2C barely broke even and REINFORCE failed to learn. Training time varied: REINFORCE was fastest (15s for 500 episodes) but learned the least; PPO and A2C took ~130-140s; DQN took ~73s.

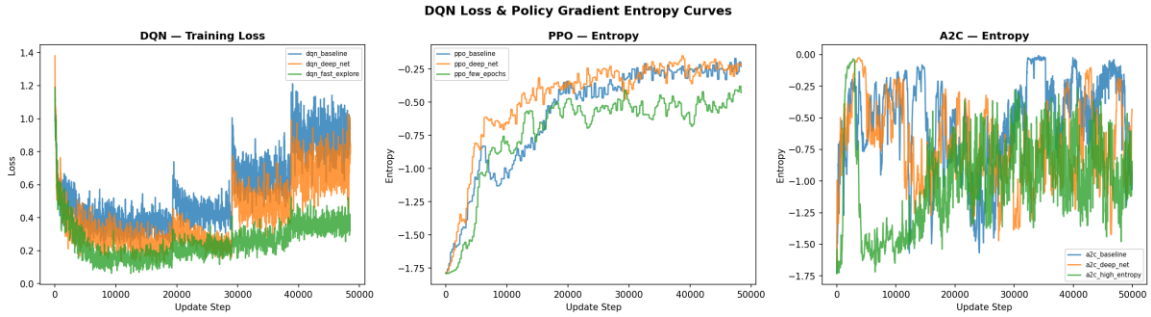
Cumulative Rewards



DQN and PPO show clear upward trends with rewards trending toward positive territory. DQN's top configs oscillate between -15 and +25 with increasing positive frequency after ~1500 episodes.

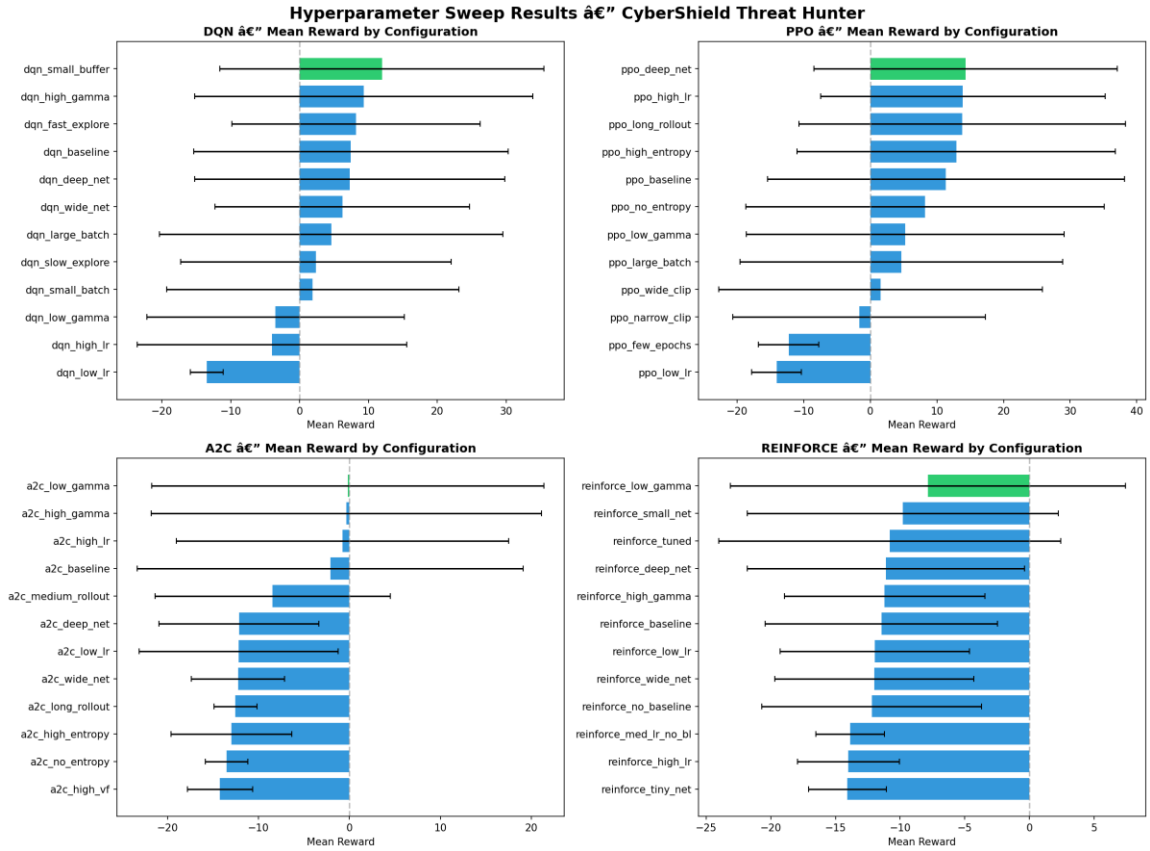
PPO's curves are smoother due to clipped updates. A2C shows more volatile training with lower overall reward levels.

Training Stability



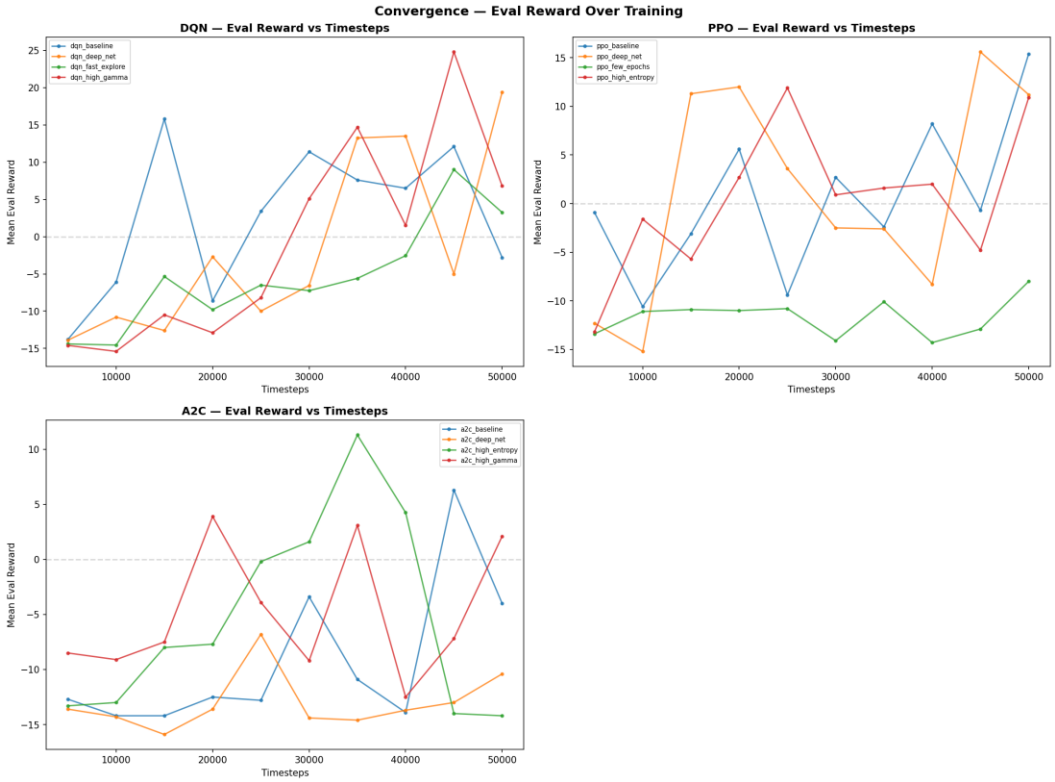
DQN training loss shows an initial spike as the replay buffer fills, followed by steady decrease and stabilization (~0.2-0.4). PPO entropy curves show gradual increase, indicating healthy exploration. A2C entropy varies more, with the baseline variant maintaining healthier entropy levels.

Hyperparameter Impact



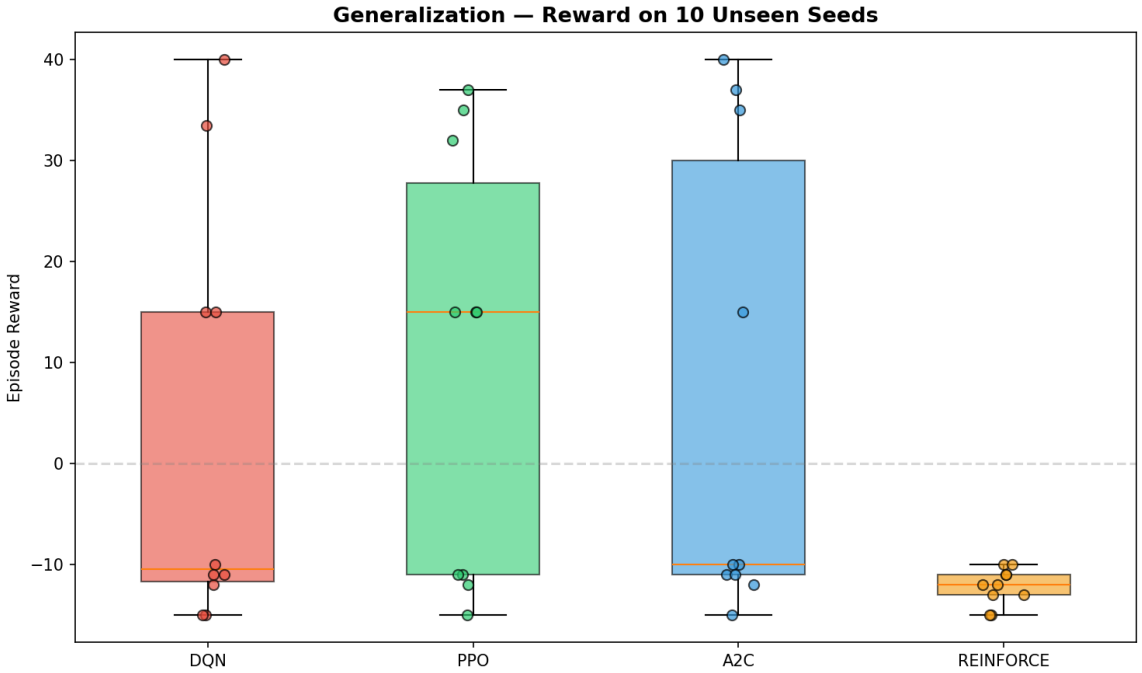
The horizontal bar charts show the impact of each hyperparameter configuration. Key patterns: very low learning rates universally hurt performance; PPO is most robust to hyperparameter variation; A2C and REINFORCE are highly sensitive; network architecture matters most for PPO.

Episodes to Converge



DQN converges around episode 1000-1500. PPO achieves stable positive rewards by episode 1500-2000. A2C does not reliably converge within 50K timesteps. REINFORCE shows no meaningful convergence within 500 episodes.

Generalization



Testing on 10 unseen seeds: PPO generalizes best (median ~15, peaks at 35-37). DQN shows bimodal behavior (either +33-40 or -12 to -15). A2C has the widest spread (some +40, many -10). REINFORCE is clustered at -10 to -15, confirming no meaningful policy was learned.

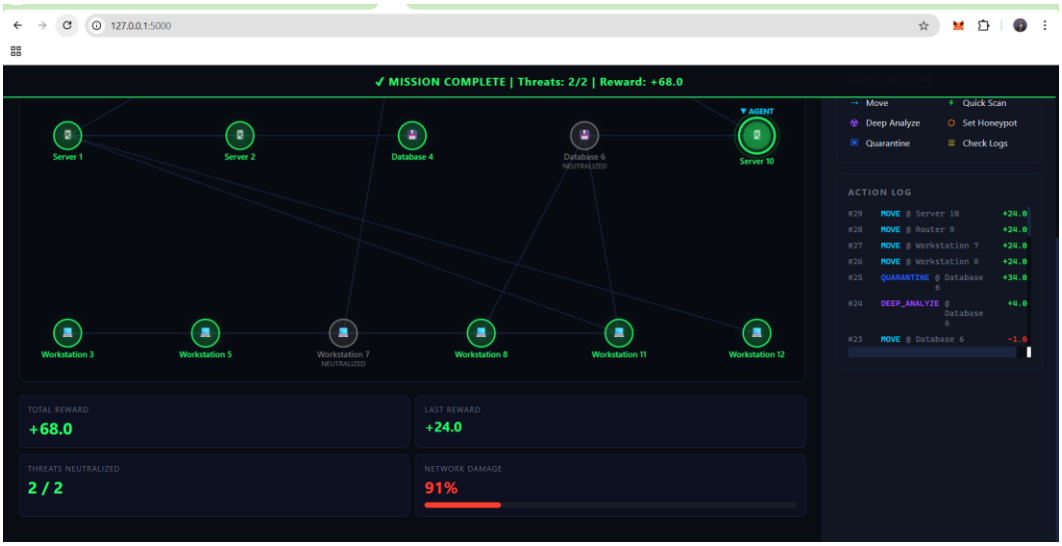
6. Conclusion and Discussion

PPO emerged as the best algorithm for CyberShield Threat Hunting, achieving +14.30 mean reward with a deep network. Its clipped objective handles the complex, partially observable environment effectively. DQN was second (+11.98), benefiting from experience replay and target networks. A2C underperformed (-0.18), limited by the 50K timestep budget. REINFORCE performed worst (-7.85) due to inherent high gradient variance.

Key insights: (1) PPO's clipping provides robustness to hyperparameter choices; (2) Deeper networks help when the algorithm can utilize the capacity; (3) Exploration strategies (entropy, epsilon, buffer size) significantly impact learning; (4) Higher gamma benefits value-based methods while lower gamma benefits high-variance policy gradient methods.

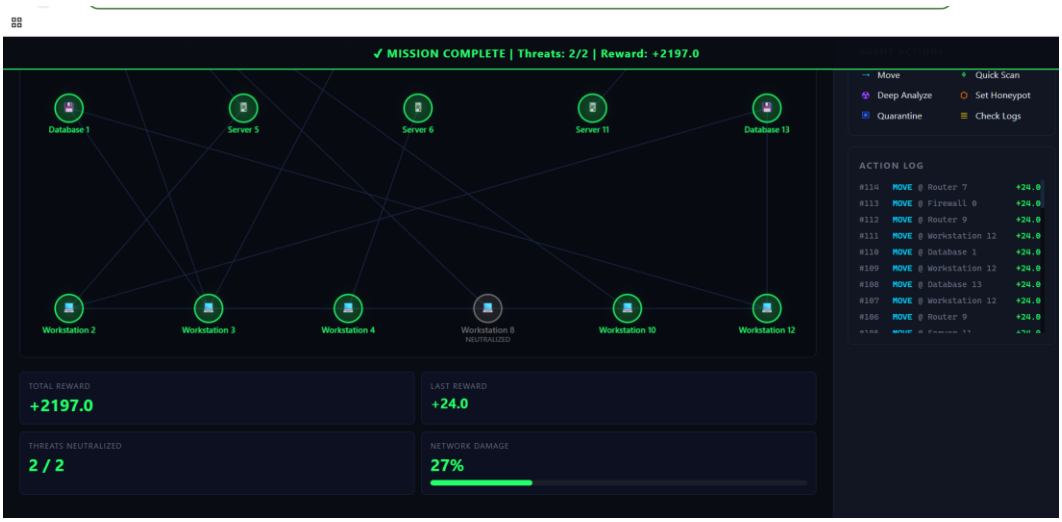
Production integration: The Flask API (api.py) serves agent predictions as JSON via REST endpoints (/predict, /step, /reset, /info), enabling integration into web or mobile cybersecurity dashboards.

Web Dashboard — Mission Complete (Threat Detection & Quarantine)



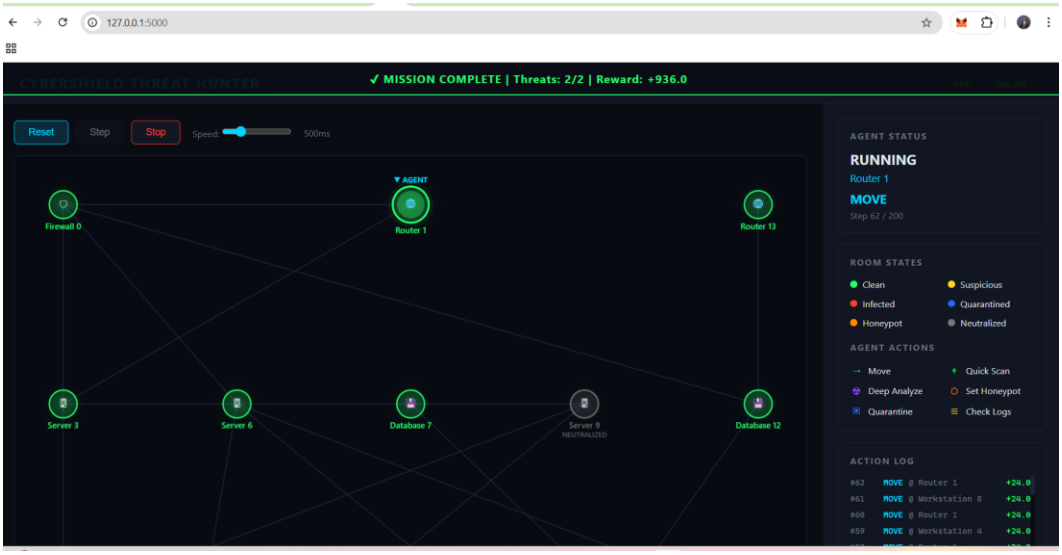
The PPO agent successfully neutralized 2/2 threats with a total reward of +68.0. The action log shows the decision sequence: MOVE to Database 6, DEEP_ANALYZE to confirm the threat (+4.0 reward), then QUARANTINE (+34.0 including completion bonus). Network damage reached 91%, demonstrating the agent completed the mission just before the critical breach threshold — showcasing urgency-driven behavior learned through the −1.0 per-step time penalty.

Web Dashboard — Multi-Episode Cumulative Performance



After multiple consecutive episodes, the cumulative reward reached +2,197.0, confirming the PPO agent consistently completes missions across different random seeds. Network damage is only 27% — significantly lower than Figure 1 — illustrating how performance varies with threat placement while remaining effective. Workstation 8 is marked NEUTRALIZED with 2/2 threats cleared.

Web Dashboard — Full Interface with Controls and Legend



The complete dashboard interface: control bar (Reset, Step, Stop, speed slider), Agent Status panel (RUNNING at Router 1, step 62/200), Room States legend (Clean, Suspicious, Infected, Quarantined, Honeypot, Neutralized), and Agent Actions legend. Server 9 is NEUTRALIZED with +936.0 total reward — demonstrating a production-ready interface suitable for cybersecurity operations center integration.

Future work: More training timesteps (200K+) for A2C; curriculum learning (fewer threats initially); recurrent layers (LSTM) for better scan history tracking; multi-agent cooperation for larger network topologies.